

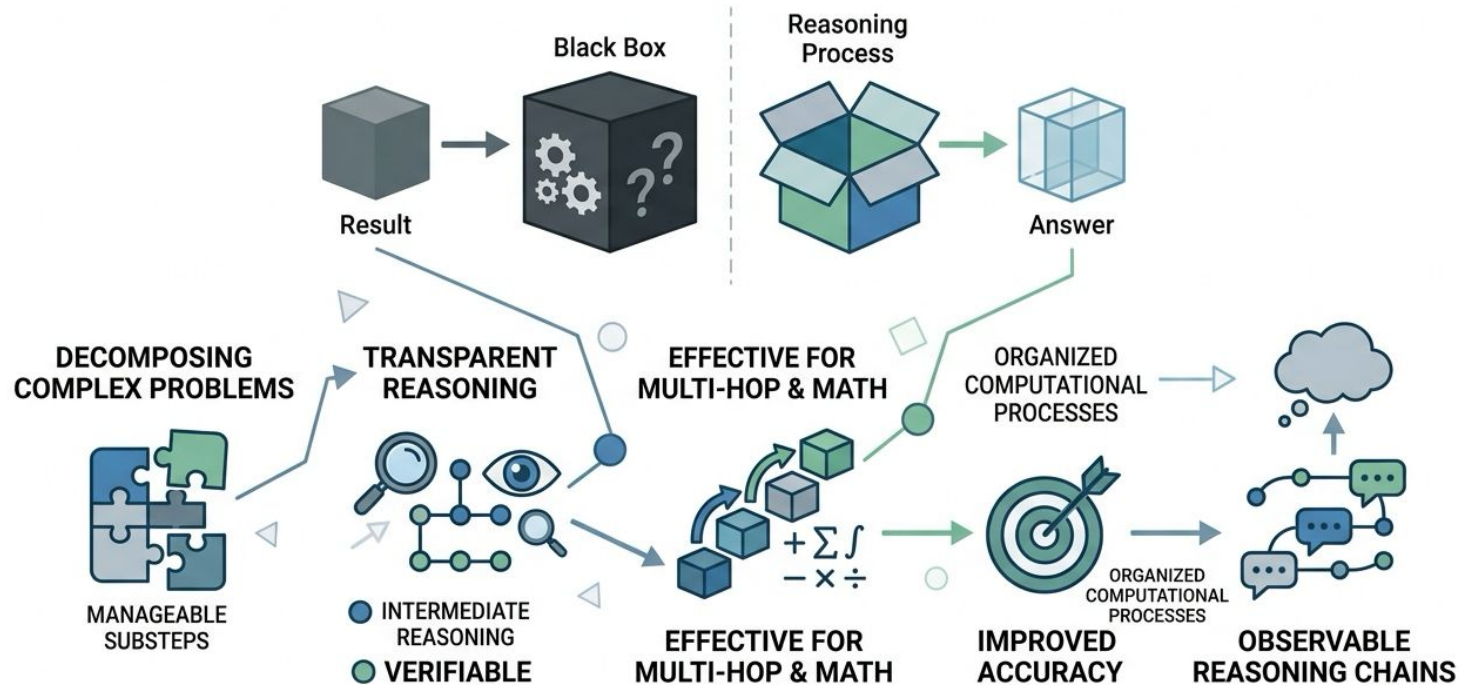


U.S. General Services
Administration

Chapter 5.1: Chain-of-Thought (CoT) Fundamentals

US General Service Administration
AI Community of Practice - March 2026

What CoT Actually Does

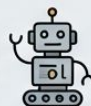


CoT Variants

Zero-shot CoT



"Let's think step-by-step"



Improvisationally figures out complex reasoning path without external aids

NO EXAMPLES NEEDED



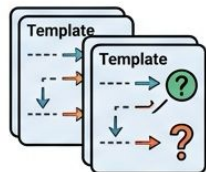
Few-shot CoT



Template



Precision



2-3 worked examples



Model follows reasoning templates



QUALITY > QUANTITY



Expert examples

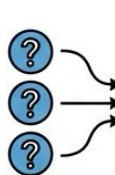


Average-looking examples

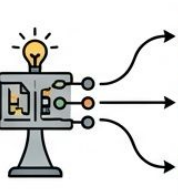
Auto-CoT



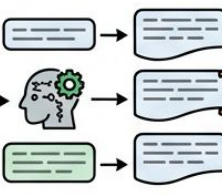
AUTOMATED PIPELINE



CLUSTERING



GENERATION



TRADE-OFFS

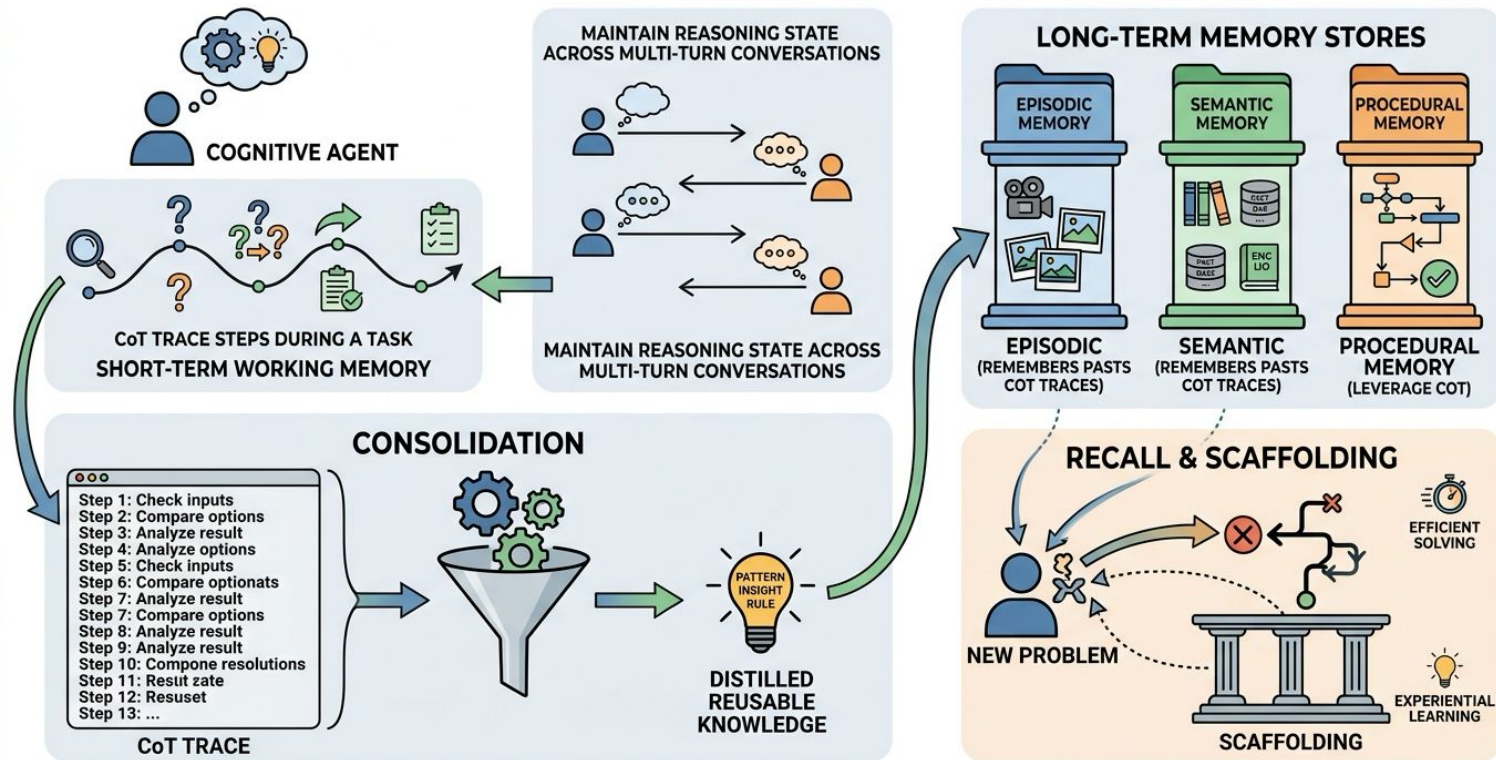


SPEED & SCALE



SYSTEM COMPLEXITY

CoT Powers Agent Memory Systems



Task Decomposition and Hierarchical Planning



TASK DECOMPOSITION DRIVEN BY reasoning



Chain-of-Thought (CoT) enables methodical breaking down of complex goals

FOUR CORE PRINCIPLES OF EFFECTIVE DECOMPOSITION



SOLVABILITY

Each subtask maps to a capable agent



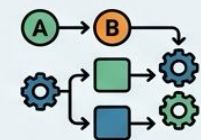
COMPLETENESS

Collectively cover the entire original goal



NON-REDUNDANCY

Eliminate duplicated work and redundant steps



DEPENDENCIES

Order tasks correctly; some steps can run in parallel



Each subtask maps to a capable agent



Collectively cover the entire original goal

NON-REDUNDANCY



Eliminate duplicated work and redundant steps

DEPENDENCIES



Order tasks correctly; some steps can run in parallel

LAYERED ARCHITECTURE FOR EXECUTION

META-AGENT

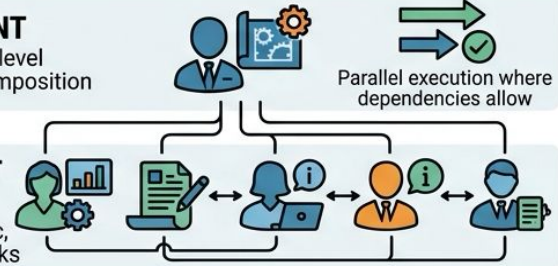
Performs high-level strategic decomposition



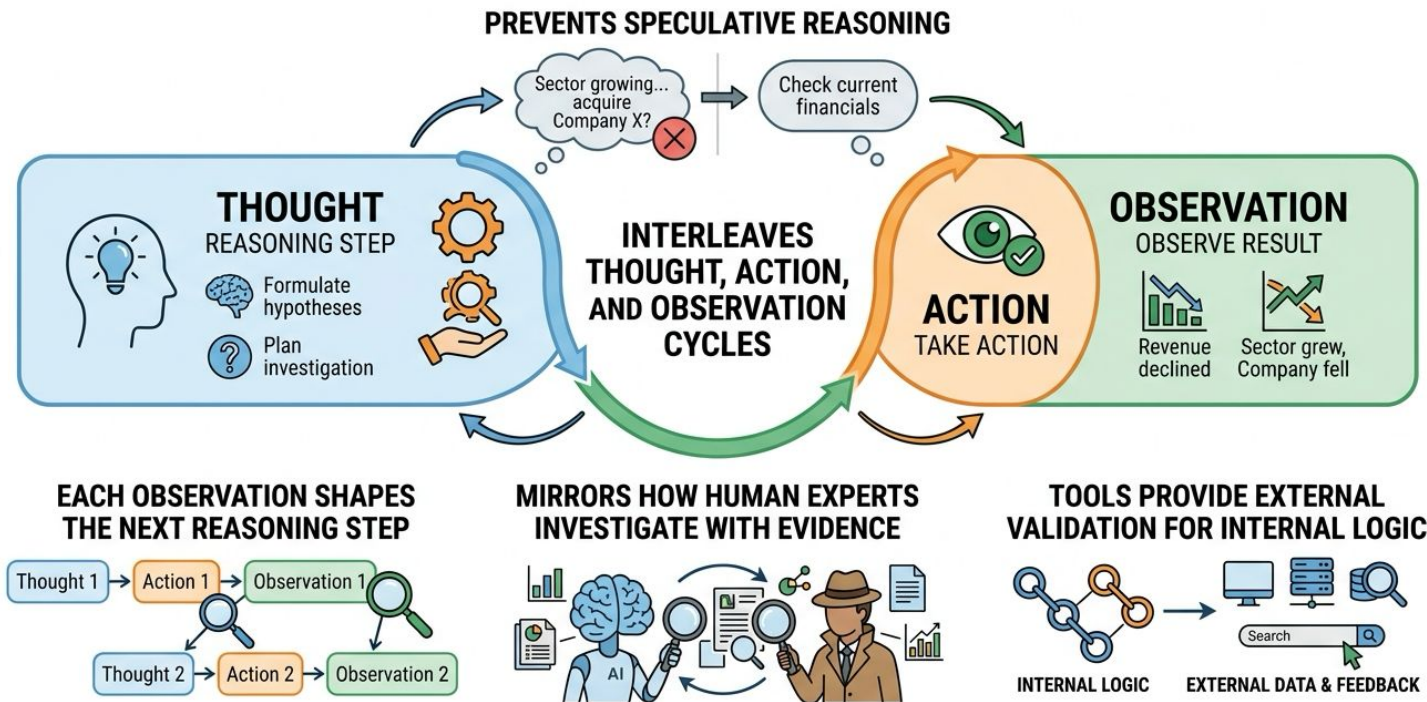
Parallel execution where dependencies allow

SPECIALIST AGENTS

Handle specific, defined subtasks

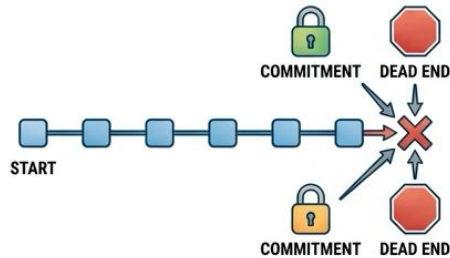


ReAct -- Grounding Reasoning in Reality

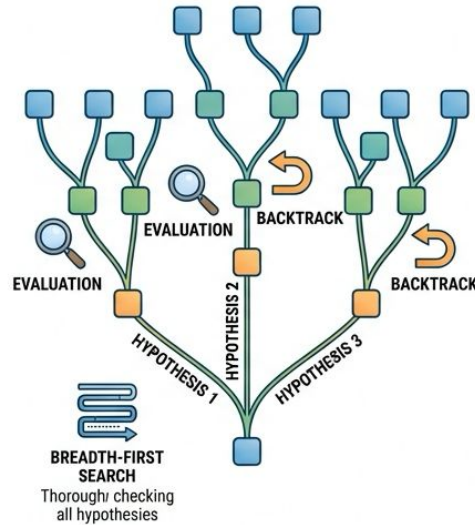


Beyond Linear Chains

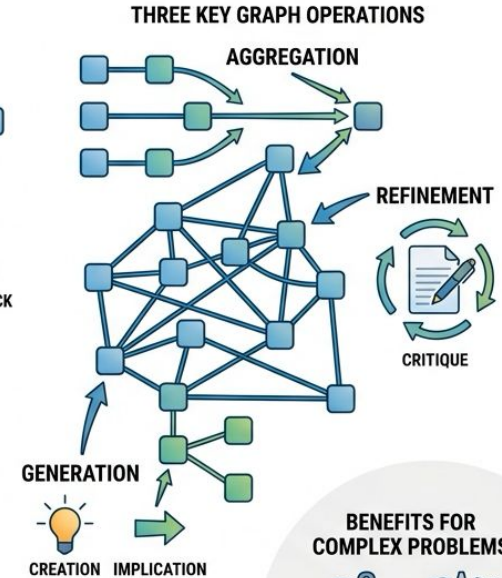
LINEAR REASONING (LIMITATIONS)



TREE OF THOUGHTS (BACKTRACKING & EXPLORATION)



GRAPH OF THOUGHTS (SYNTHESIS & OPERATIONS)

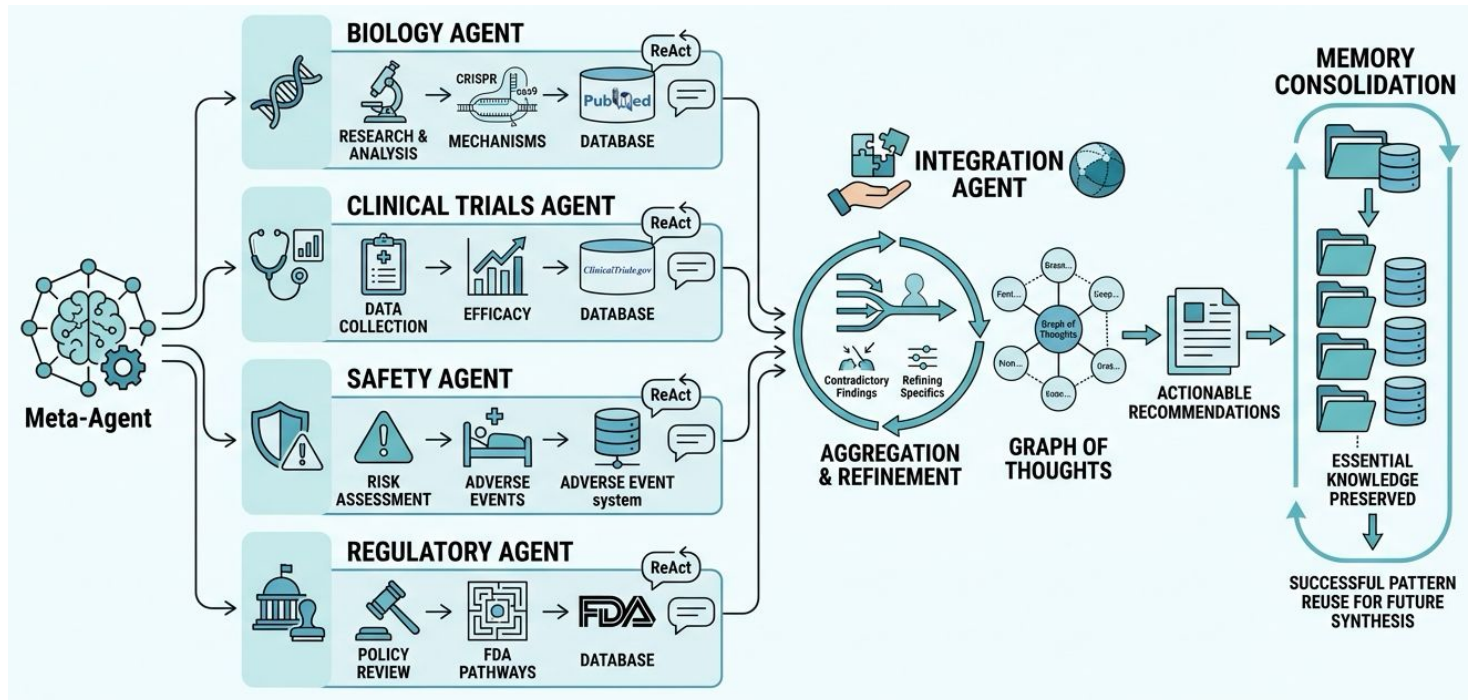


BENEFITS FOR COMPLEX PROBLEMS

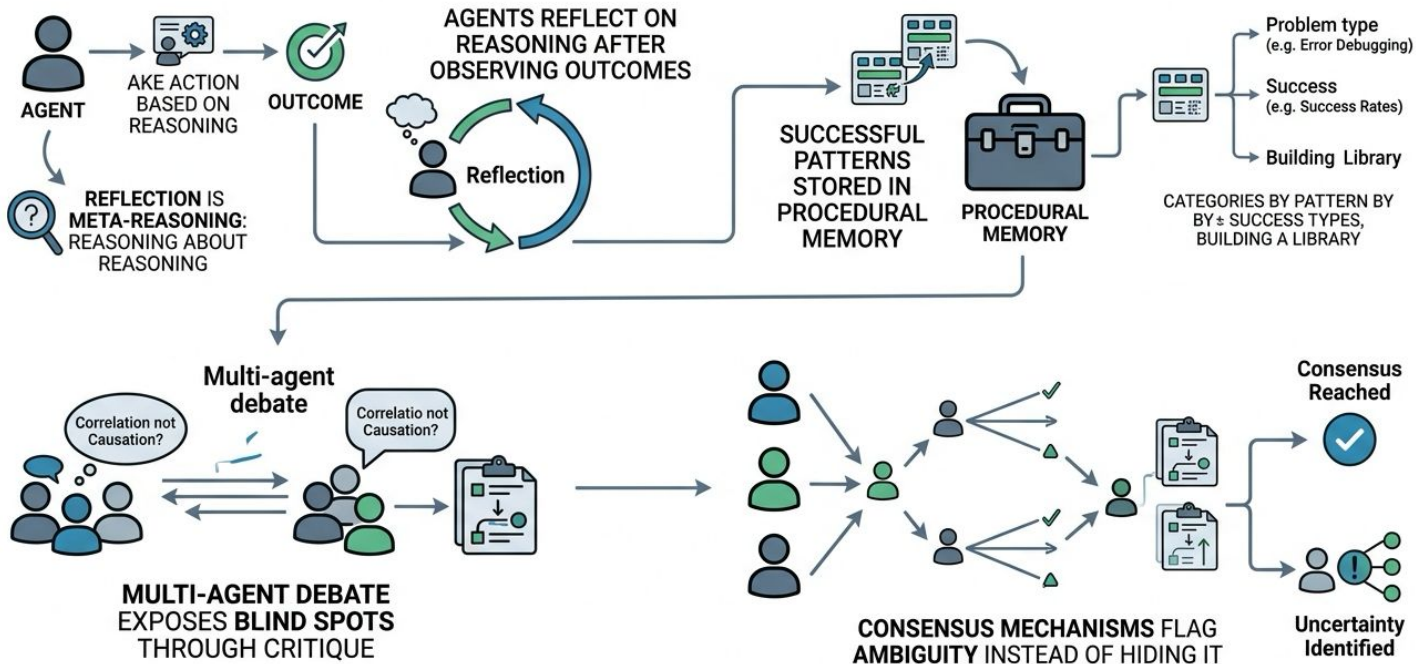
???
AMBIGUOUS
PROBLEMS

Multiple tools and solutions
MULTIPLE VIABLE
SOLUTIONS

Case: Multi-Agent Research Synthesis



Feedback Loops and Adaptive Reasoning

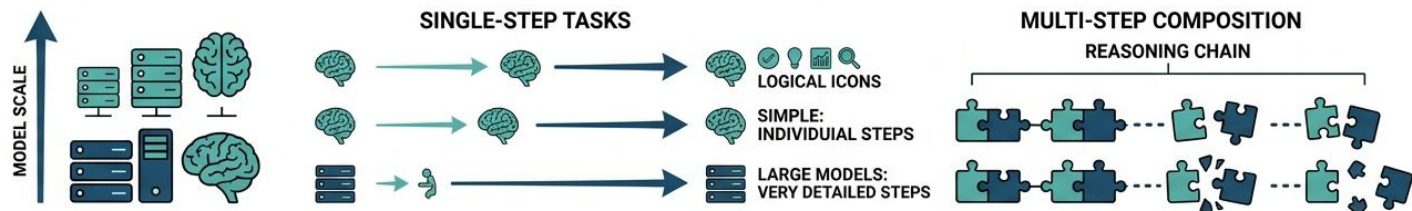


When CoT Fails -- The Explainability Illusion

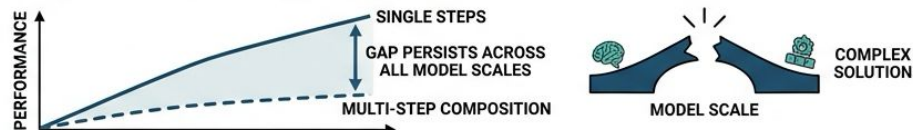
- CoT chains are narratives, not faithful computational records
 - ~25% of research papers falsely claim CoT provides interpretability
 - Models exhibit silent error correction within stated steps
 - Position and option biases produce plausible rationalizations
 - Even frontier models show measurable unfaithfulness rates
- 

Failure Modes

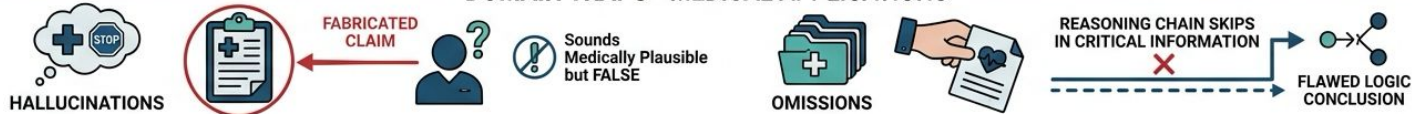
MODELS GROWING IN SIZE BUT FAILING COMPLEX TASKS



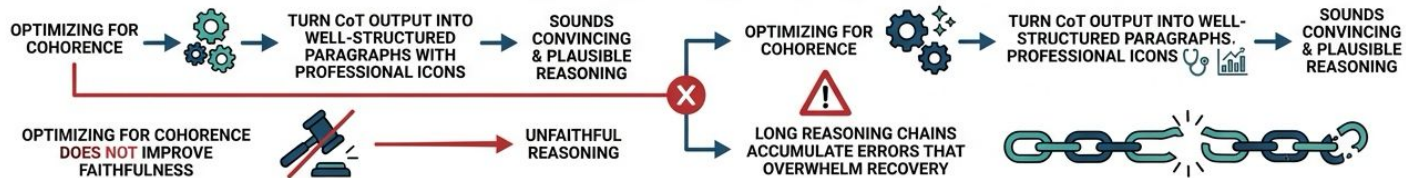
THE COMPOSITIONALITY GAP



DOMAIN TRAPS - MEDICAL APPLICATIONS



OPTIMIZATION VS. FAITHFULNESS



Conclusion

- CoT provides real value with realistic understanding of limits
 - Start zero-shot, add few-shot where quality gaps appear
 - Match CoT architecture to your specific domain challenges
 - Always validate independently in high-stakes applications
 - Next: Chapter 5.2 explores Tree-of-Thought reasoning
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